

# The Relationship Between Song Energy and Popularity Across Genres

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This study investigates the relationship between the perceived energy of a song and its popularity across multiple musical genres. We analyzed 25,736 songs from six genres with nonzero popularity. Distributions of energy and popularity were summarized, and Pearson's correlation test was conducted to test their association. This was done for the dataset as a whole, and across genres. A Simple linear regression model was fit to quantify the correlation between energy and popularity. Multiple linear regression model, including an energy \* genre interaction, was fit to evaluate whether this relationship varied by genre. Results show a statistically significant but weak, negative, and linear association between energy and popularity, which however showed energy explained little variation in popularity. Across genres, this relationship did not meaningfully vary, indicating that energy alone is a limited predictor of song popularity. Future research can test different confounding variables for a better understanding of popularity of a song.

## Introduction and Background

Music platforms like Spotify allow researchers access to measured audio features such as energy, danceability, and valence that shape the how a listener experiences the song. Prior research has shown that audio features can be used to explain trends in popular music over time. (Mauch et al. 2015) . Other analyses have found that energy is linked to emotional response and listener engagement, but the relationship with popularity is more nuanced. (North et al. 2018) . It is clear that audio features are valuable for studying what makes a hit, but the role of energy in predicting popularity, and how it is relevant in each different genre, remains to be explored. In this research we investigate the research question. What is the relationship between a song's energy level and it's popularity, and does it vary by genre?

## Study Design and Data collection

For this observational study, we used 'The Spotify Songs Dataset' which was originally extracted directly from Spotify directly using Web API by tidyuesday community (TidyTuesday

2020). It contains 25,736 individual Songs dating from 1960 to 2020. Song's were collected from playlists curated by users, with song popularities ranging from 0 to 100. Each song in the dataset includes both its popularity and audio features.

## **Methods**

### **Data Preparation**

All variables of interest were screened and visualized to verify accuracy and valid ranges. However, the `track_popularity` variable was zero-inflated: many songs had a recorded popularity value of zero, indicating no measurable listener engagement. Tracks with zero popularity were excluded. Duplicate records were removed by keeping a single record per unique `track_id`.

Our variables of interest include `popularity_nonzero`, a quantitative variable that ranges from 1 to 100 where higher is more popular. This is going to serve as our response variable. For explanatory variables, we have `energy`, another quantitative variable that measures perceptual intensity/activity from the song. Measured on a scale of 0 to 1, 0 is low energy and 1 is higher energy. Both of these variables are generated by Spotify's own algorithm. We are going to test the variable `playlist_genre` as a moderator. This nominal categorical variable divides the songs into six distinct genres, namely: Pop, Rap, EDM, R&B, Rock, and Latin.

### **Statistical Analysis**

We conducted exploratory data analysis using barcharts, histograms, and violin plots to describe the distributions for our variables of interest: `energy`, `popularity_nonzero`, and `playlist_genre`. Scatter plots with lines were used to explore the potential relationship between our variables. Pearson's product-moment correlation test was used to examine and quantify the relationship between a song's energy and popularity, as both of these are quantitative variables. Next, we used genre as a moderator to do a stratified version of our correlation test, to see how this relationship varies across different genres.

For regression analysis, we built multiple linear regression models to see the association between a song's energy and it's popularity, and also to see this association across different genres. We started with a simple linear regression model by fitting popularity and energy to see the overall association between the two variables. Next, we added genre to test as a categorical moderator to see whether this relationship is consistent or changes across the genres

## Results

### Descriptive Statistics

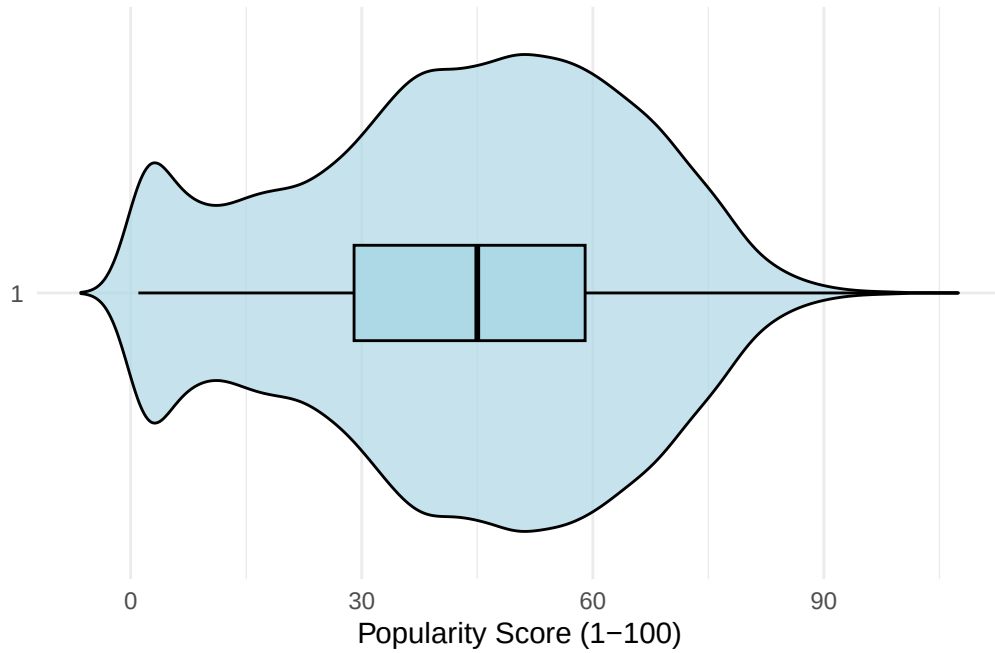


Figure 1: Distribution of popularity\_nonzero

The distribution for popularity\_nonzero, as shown in Figure 1, is roughly symmetric and slightly bimodal, with one peak in the late 30s–40s and another smaller bump around the early 50s. The mean is 43.3 and the median is 45, confirming little skew. The standard deviation is about 21, showing a wide spread across the scale. Half of the observations lie between 29 and 59, indicating that most values cluster in the middle, with fewer at the very low and very high ends.

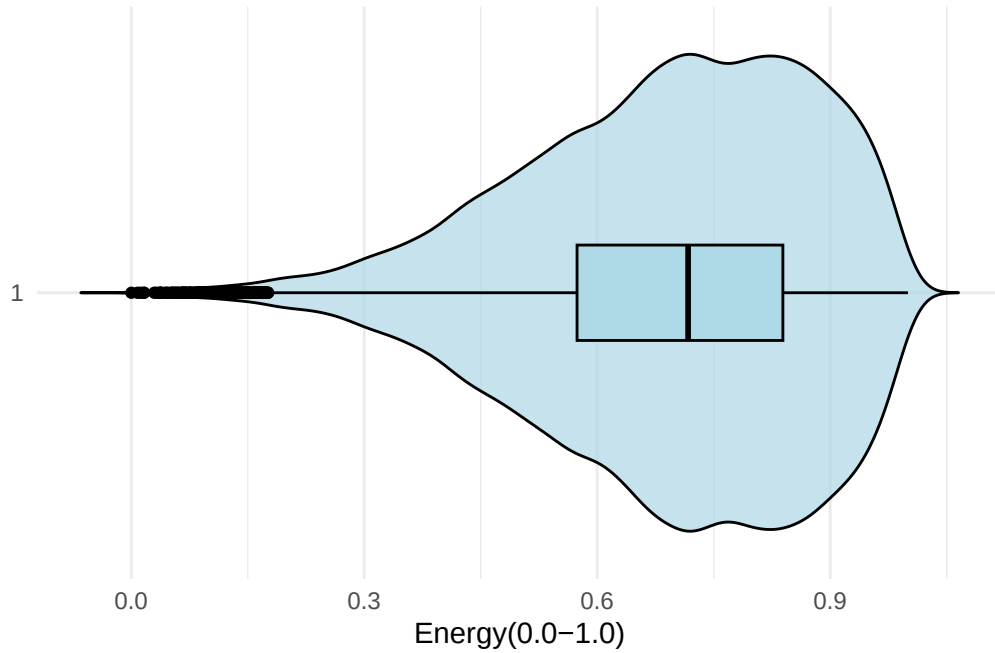


Figure 2: Distribution of energy

From Figure 2, we see that energy ranges from 0.0002 to 1.0 on Spotify's 0.0–1.0 scale. The mean is 0.69 and the median is 0.72, with half of tracks between 0.57 and 0.84. The distribution is slightly left-skewed, showing that most tracks in the dataset are moderately to highly energetic, with relatively few low-energy songs.

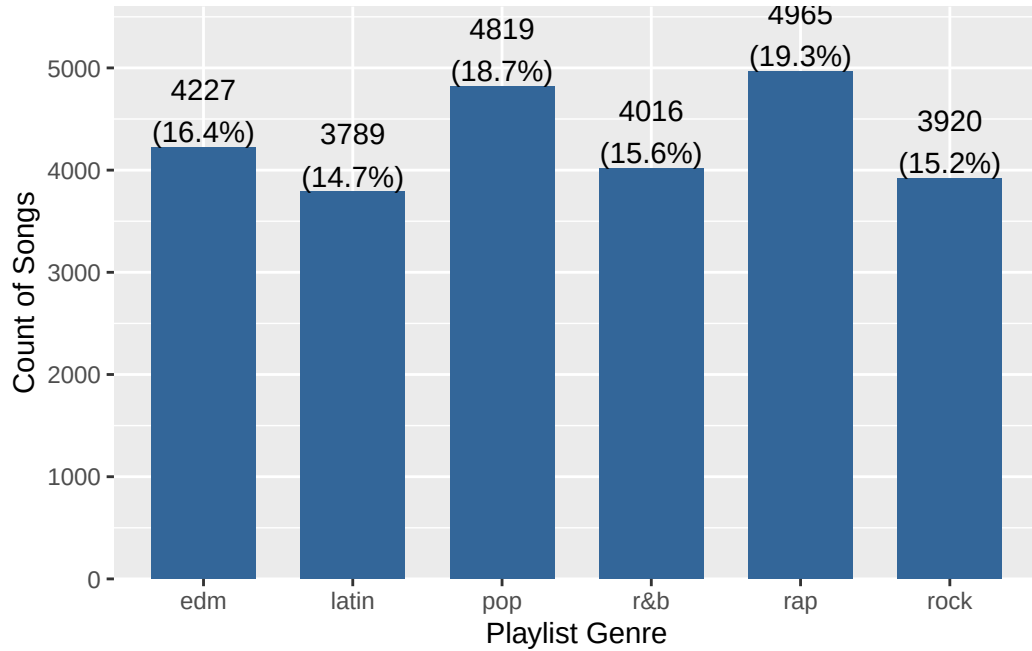


Figure 3: Box Plot of genre

In Figure 3 we see that the songs are divided into their genres with a fairly even distribution, and minor variations between the genres. The genre rap has the largest number of songs with a 19.1% share ( $N = 4965$ ), while latin has the smallest share of 14.6% ( $N = 3789$ ). The dataset gives us a diverse collection of genres without a large concentration in a specific genre.

Table 1

| Characteristic                  | N = 25,736 <sup>1</sup> |
|---------------------------------|-------------------------|
| <b>Playlist Genre</b>           |                         |
| edm                             | 4,227 / 25,736 (16%)    |
| latin                           | 3,789 / 25,736 (15%)    |
| pop                             | 4,819 / 25,736 (19%)    |
| r&b                             | 4,016 / 25,736 (16%)    |
| rap                             | 4,965 / 25,736 (19%)    |
| rock                            | 3,920 / 25,736 (15%)    |
| <b>Track Popularity (1-100)</b> | 43 (21)                 |

| Characteristic   | N = 25,736 <sup>1</sup> |
|------------------|-------------------------|
| Energy (0.0-1.0) | 0.69 (0.18)             |

<sup>1</sup>n / N (%); Mean (SD)

Table 1 summarizes all of our findings about the variables that are of interest to us.

## Bivariate Analysis

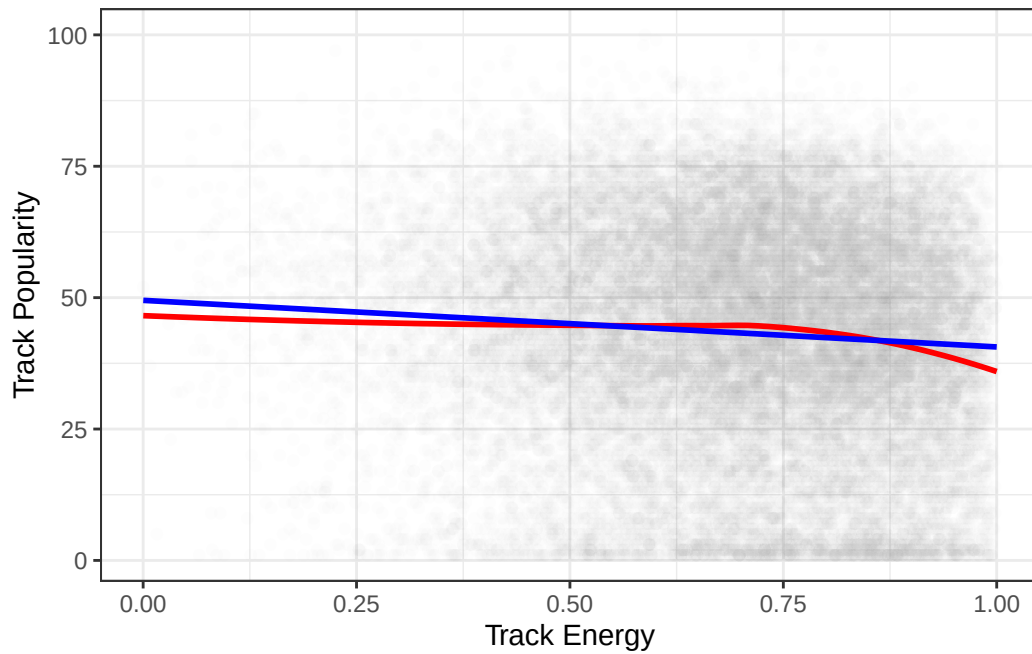


Figure 4: Correlation between energy and popularity

Pearson's product-moment correlation

```
data: clean$energy and clean$popularity_nonzero
t = -12.425, df = 25734, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.08935732 -0.06506808
sample estimates:
cor
```

-0.07722416

In Figure 4, the blue is our linear model line, and the red one shows our actual variables' correlation. Combined with the Pearson's product-moment correlation test we see that there was a statistically significant but very weak, negative correlation between song energy and song popularity ( $r = -0.08$ , 95% CI:  $-0.089$  to  $-0.065$ ,  $p < .0001$ ). These results suggest that less than 1% of the variance in popularity can be explained by a song's energy level.

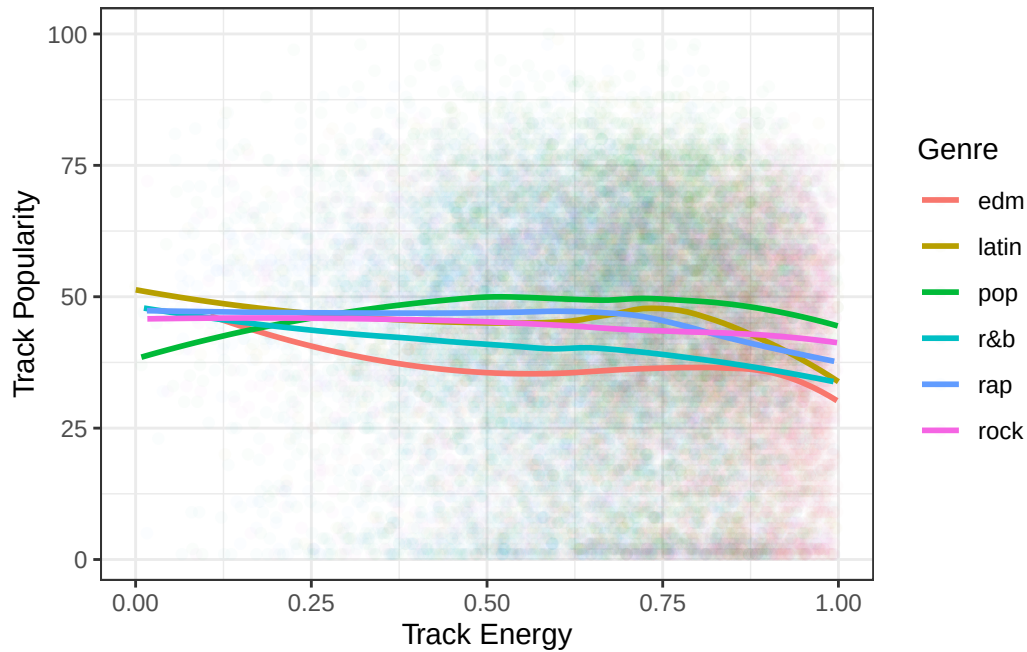


Figure 5

Our second correlation test between the same variables, but divided by genres, as shown in Figure 5, again we see different association for all of them. It differs in its strength, direction, and pattern for each different genre.

### Regression Modelling

As we see in Table 2, when we fit our linear regression model for energy and popularity\_nonzero, we find that for a song with an energy value of 0, the predicted popularity score is 49.49 (95% CI: 48.48, 50.49). Each 1 unit increase in energy is associated with a significant decrease of 8.87 (95% CI:  $-10.27$ ,  $-7.47$ ) points in popularity ( $p < .001$ ). Energy explains only 0.6% of the variation in song popularity.

Table 2

| Characteristic | Beta   | 95% CI         | p-value |
|----------------|--------|----------------|---------|
| (Intercept)    | 49.49  | 48.48, 50.49   | <0.001  |
| energy         | -8.867 | -10.27, -7.468 | <0.001  |

Abbreviation: CI = Confidence Interval

$R^2 = 0.006$

Table 3

| Characteristic          | Beta   | 95% CI      | p-value |
|-------------------------|--------|-------------|---------|
| (Intercept)             | 39     | 36, 43      | <0.001  |
| energy                  | -5.1   | -9.6, -0.59 | 0.026   |
| playlist_genre          |        |             |         |
| edm                     | —      | —           |         |
| latin                   | 10     | 5.7, 15     | <0.001  |
| pop                     | 10     | 5.8, 15     | <0.001  |
| r&b                     | 6.8    | 2.6, 11     | 0.002   |
| rap                     | 12     | 8.2, 17     | <0.001  |
| rock                    | 8.5    | 4.1, 13     | <0.001  |
| energy * playlist_genre |        |             |         |
| energy * latin          | -1.6   | -7.8, 4.5   | 0.6     |
| energy * pop            | 3.9    | -1.7, 9.5   | 0.2     |
| energy * r&b            | -5.2   | -11, 0.49   | 0.073   |
| energy * rap            | -5.0   | -11, 0.58   | 0.079   |
| energy * rock           | -0.90  | -6.4, 4.6   | 0.7     |
| Adjusted $R^2$          | 0.046  |             |         |
| No. Obs.                | 25,736 |             |         |

Abbreviation: CI = Confidence Interval



Fitting our multiple linear regression model, in Table 3 with an interaction between energy and playlist\_genre, we find that for a song in the reference group (EDM) with an energy value of 0, the predicted popularity score is 39 (95% CI: 36, 43,  $p < .001$ ). Each 1 unit increase in energy is associated with a significant decrease of 5.1 points in popularity for EDM songs (95% CI: -9.6, -0.59,  $p = 0.026$ ). Compared to EDM (holding energy at 0), Latin songs are 10 points higher on average (95% CI: 5.7, 15,  $p < .001$ ), Pop songs are 10 points higher (95% CI: 5.8, 15,  $p < .001$ ), R&B songs are 6.8 points higher (95% CI: 2.6, 11,  $p = 0.002$ ), Rap songs are 12 points higher (95% CI: 8.2, 17,  $p < .001$ ), and Rock songs are 8.5 points higher (95% CI: 4.1, 13,  $p < .001$ ).

## Discussion, Conclusion

From our Table 2, we can conclude that there is significant evidence to suggest that energy is negatively associated with popularity. This means that an increase in songs energy, on average, decreases its popularity.

When we test genre as a moderator in Table 3, we see that the p values for most of our interactions indicate they are not statistically significant. The interaction terms (energy\*genre) quantify how the energy x popularity slope differs from our reference group (EDM) within each genre. R&B and Rap show larger negative associations compared to the reference group, although with some uncertainty. (energy  $\times$  R&B = -5.2, 95% CI: -11, 0.49; energy  $\times$  rap = -5.0, 95% CI: -11, 0.58)

It is also important to note the adjusted  $R^2$  values for both of our models. 0.6% and 4.6% are extremely low values, which demonstrate this is not the best model fit.

Energy plays a small role in predicting a songs popularity. There are likely many other factors which have a more meaningful hand in dictating a songs popularity, or a combination of multiple different factors. This is consistent with previous studies. (See (North et al. 2018))

Music artists can use the information that energy has small but negative association with popularity, and to make their music more commercially friendly, by keeping the energy in check. Playlist curators can prioritize tracks with balanced energy levels over higher ones when making playlists as these will tend to perform better. Future research could explore other variables like valence, danceability, and see how they interact with energy and what role do these play in predicting a songs popularity.

However, all of these findings are limited by some factors. Our dataset was only covering songs from Spotify, so these findings may not be representative of music not found on spotify, and is from different regions, or genres outside of the 6 we covered. It is also important to note the adjusted  $R^2$  values for both of our models. 0.6% and 4.6% are extremely low values, which demonstrate this is not the best model fit. This is an observational study, so we cannot establish causal relationships, only associations. Other factors not included like marketing,

prior artist popularity also may play a key role and are not used in our model, or even present in our dataset.

Future research could explore other variables like valence, danceability, and see how they interact with energy and what role do these play in predicting a songs popularity.

## References

- Mauch, Matthias, Robert M. MacCallum, Mark Levy, and Armand M. Leroi. 2015. “The Evolution of Popular Music: USA 1960–2010.” *Royal Society Open Science* 2 (5): 150081. <https://doi.org/10.1098/rsos.150081>.
- North, Adrian C., Amanda E. Krause, Lorraine P. Sheridan, and David Ritchie. 2018. “Energy, Popularity, and the Circumplex: A Computerized Analysis of Emotion in 143,353 Musical Pieces.” *Empirical Studies of the Arts* 36 (2): 127–61. <https://doi.org/10.1177/0276237417704339>.
- TidyTuesday. 2020. “The Spotify Songs Dataset.” [https://github.com/rfordatascience/tidyuesday/blob/main/data/2020/2020-01-21/spotify\\_songs.csv](https://github.com/rfordatascience/tidyuesday/blob/main/data/2020/2020-01-21/spotify_songs.csv).

## Appendix

This following code describes our methodology, from cleaning our raw dataset to generating plots.

Libraries, and reading in data

```
library(tidyverse)
library(gtsummary)
library(sjPlot)
library(ggplot2)
library(dplyr)
library(ggpubr)
library(ggdist)
library(performance)

# reading the raw data set
raw <- readr::read_csv("data/spotify_songs.csv")
```

Cleaning

```
raw_de <- raw %>% group_by(track_id) %>%
  filter(track_popularity == max(track_popularity)) %>% slice(1) %>% ungroup()

clean <- raw_de %>%
  mutate(popularity_nonzero = ifelse(track_popularity > 0, track_popularity, NA)) %>%
  filter(!is.na(popularity_nonzero))
```

Distribution of popularity\_nonzero

```
ggviolin(clean$popularity_nonzero, add = "boxplot", fill = "lightblue", alpha = 0.7) +
  coord_flip() +
  labs(x = "", y = "Popularity Score (1-100)") +
  theme_minimal()
```

Distribution of energy

```
ggviolin(clean$energy, add = "boxplot",
  fill = "lightblue", alpha = 0.7) +
  coord_flip() +
  labs(
    x = "", y = "Energy(0.0-1.0)") +
  theme_minimal()
```

Box Plot of Genre

```
plot_frq(clean$playlist_genre) + labs(x = "Playlist Genre", y = "Count of Songs")
```

Summary Statistics

```
library(flextable)
clean %>%
  select(playlist_genre, popularity_nonzero, energy) %>%

  tbl_summary(

    statistic = list(
      all_continuous() ~ "{mean} ({sd})",          # For quant: Mean (SD)
      all_categorical() ~ "{n} / {N} ({p}%)"        # For cat: n / N (%)
    ),
```

```

label = list(
  playlist_genre ~ "Playlist Genre",
  popularity_nonzero ~ "Track Popularity (1-100)",
  energy ~ "Energy (0.0-1.0)"
),

) %>%
bold_labels() %>%
as_flex_table() %>%
autofit()

```

Correlation b/w energy and popularity

```

ggplot(clean, aes(x = energy, y = popularity_nonzero)) +
  geom_point(alpha = 0.01, color = "grey50") +
  geom_smooth(method = "loess", se = FALSE, color = "red") +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    x = "Track Energy",
    y = "Track Popularity"
  ) +
  theme_bw()

```

Corr Test

```

cor.test(clean$energy, clean$popularity_nonzero)

```

Correlation b/w energy and popularity across genres

```

ggplot(clean, aes(x = energy, y = popularity_nonzero, color = playlist_genre)) +
  geom_point(alpha = 0.02) +
  geom_smooth(method = "loess", se = FALSE, linewidth = 1) +
  labs(
    x = "Track Energy",
    y = "Track Popularity",
    color = "Genre"
  ) +
  theme_bw() +
  theme(
    legend.position = "right",
    plot.title = element_text(hjust = 0.5)
  )

```

Popularity ~ Energy Linear Model

```
energy.pop.model <- lm(popularity_nonzero ~ energy, data=clean)
energy.pop.model %>% tbl_regression(intercept=TRUE,
                                     estimate_fun = purrr::partial(style_ratio, digits = 3)) |>
  add_glance_source_note(include = c(r.squared))
```

Popularity ~ Energy \* Genre Linear Model

```
energy.genre.int.model <- lm(popularity_nonzero ~ energy * playlist_genre, data = clean)

energy.genre.int.model %>%
tbl_regression(intercept = TRUE) %>%
add_glance_table(include = c(adj.r.squared, nobs))
```